



Go for an Open Source Database in a Lab

The author presents a case for the use of open source software in a contract analysis lab, based on his own experience.

This article seeks to answer the question, “Does a contract analysis laboratory need big, commercial software to manage the workflow shown in Figure 1?” The answer is a very short and emphatic, “No” And this is because of FOSS.

The workflow

A contract analysis laboratory operates in the fields of analytical chemistry, microbiology and industrial hygiene. It provides its customers chemical and microbiological analysis, physically measures (noise, vibrations, etc), and offers various types of technical consultancy related to water, soil, air, workplace hygiene, food, textiles, home products and industrial products. In other words, it is closely associated with the environment and a lot of other things.

The workflow shown in Figure 1 is a bit simplified and has the following explanation. An analysis request comes from a customer, and one or more samples are sent to the lab. When the sample arrives in the lab, it must be registered and stored before the next step is taken. Then the analysis is carried out using traditional methods like titration, etc. or instrumental methods (chromatography, spectroscopy, etc). This step requires materials,

reagents and instruments supplied from some suppliers. The results of each measurement are written in a *sample notebook* (either printed on paper sheets or recorded digitally). When the analysis is completed, the results are reviewed by one or more persons. This step depends on the company's policy about quality assurance and can also be called the validation (primary and/or secondary) process. Last of all, an *analysis report* is written with all the results and comments, and is signed - usually by one or two people. The signing step depends on local laws. The person who signs the report is usually a biologist, a chemist or both. Other professionals can also analyse and sign, depending on what the law of the country requires and the report's contents. The report can be sent to the customer on paper or in a digital format that's electronically signed.

The toolbox

In my work experience, I have encountered only one company that uses GNU Linux as a desktop system (which had been installed by me!). So, I think that to have something that is usable by all, FOSS on Windows should be considered at least from a desktop point of view. For this application, the following software has been chosen: